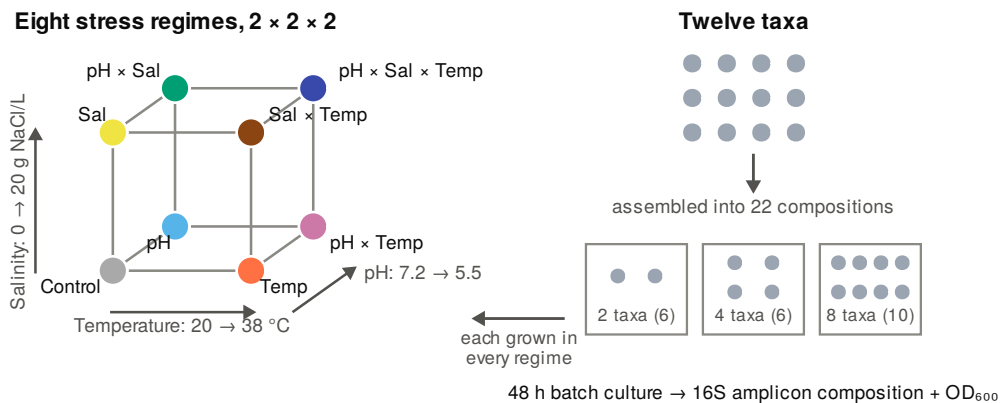
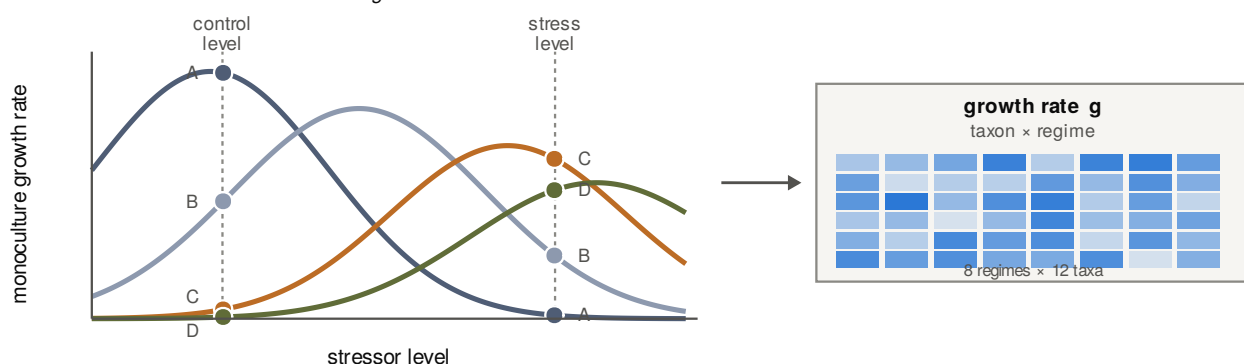


a

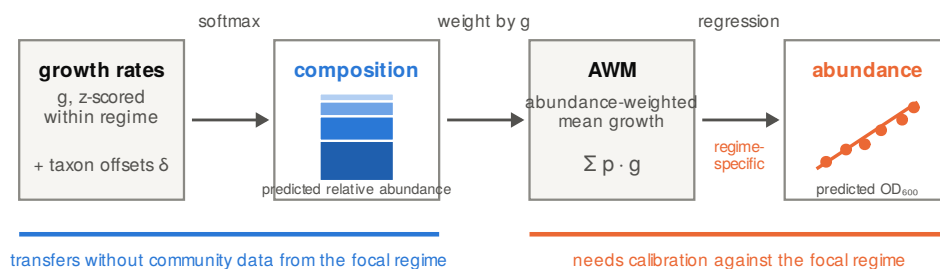


b

Reaction norms measured once per taxon (Carmichael et al. 2025)
rank order changes between levels



c



d

Held-out test

Unseen communities	Unseen taxa	Unseen stress regimes	Combinations from single stressors
22 compositions, blocked 5-fold	each taxon absent from every fitted community	each regime absent from fitting	fitted on single stressors only
composition ✓	composition ✓	composition ✓	composition ✓
abundance ✓	abundance —	abundance ✗	abundance ✗
✓ transfers	✗ requires focal-regime calibration	✗ requires focal-regime calibration	— not assessed